KAPASA MAKASA UNIVERSITY

SCHOOL OF APPLIED SCIENCE AND OPEN LEARNING

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

MOBILE DEVICE SECURITY AND FORENSICS
(CYS 453)

CONTINUOUS ASSESSMENT TEST (CAT)

DATE: 25th SEPTEMBER 2026 TIME: 10:00 – 12:00

INSTRUCTIONS

1. Write your Student Identification Number on the answer sheet provided.
2. Answer ALL FOUR (4) questions in this test.
3. This test carries a total of 50 marks.
4. Illegible handwriting will not be marked.

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Question 1 [10 Marks]

During an investigation into suspected corporate espionage, an Android device belonging to a person of interest is lawfully seized. Before deciding which acquisition method to use, the examiner checks the device for indicators of prior modification. The terminal session is shown in Figure 1.0 below.

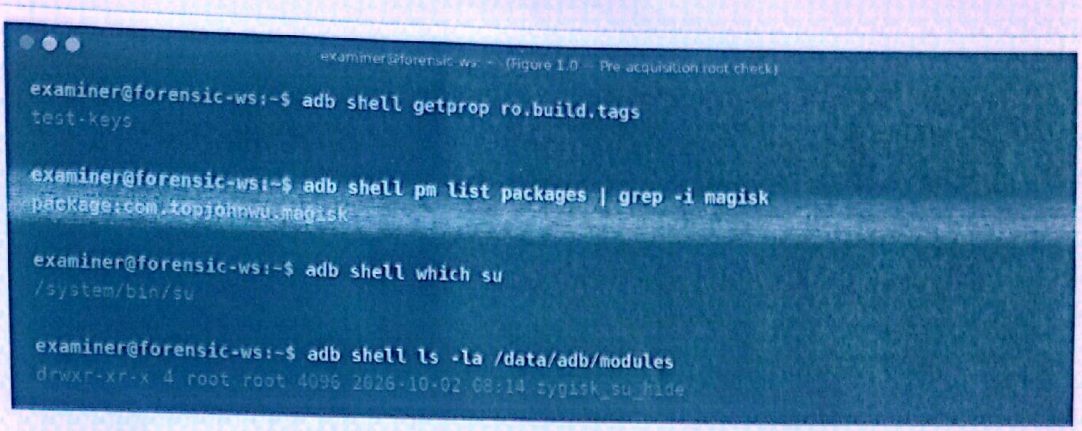


Figure 1.0

(a) Define the following terms:

(i) Root detection **[1]**

(ii) Zygisk module **[1]**

(b) Using Figure 1.0, explain what each of the four commands reveals about the device's state, and explain why an examiner would check for this before beginning acquisition. **[4]**

(c) Given that the device is confirmed to be rooted, evaluate which acquisition tier (logical, file-system, or physical) is now both possible and most appropriate for this case, and justify your answer. **[4]**

Question 2 [8 Marks]

A budgeting mobile application, com.budgetapp, exposes a content provider used to store personal financial notes. During a security assessment, a tester queries the provider directly using ADB. The terminal session is shown in Figure 2.0 below.

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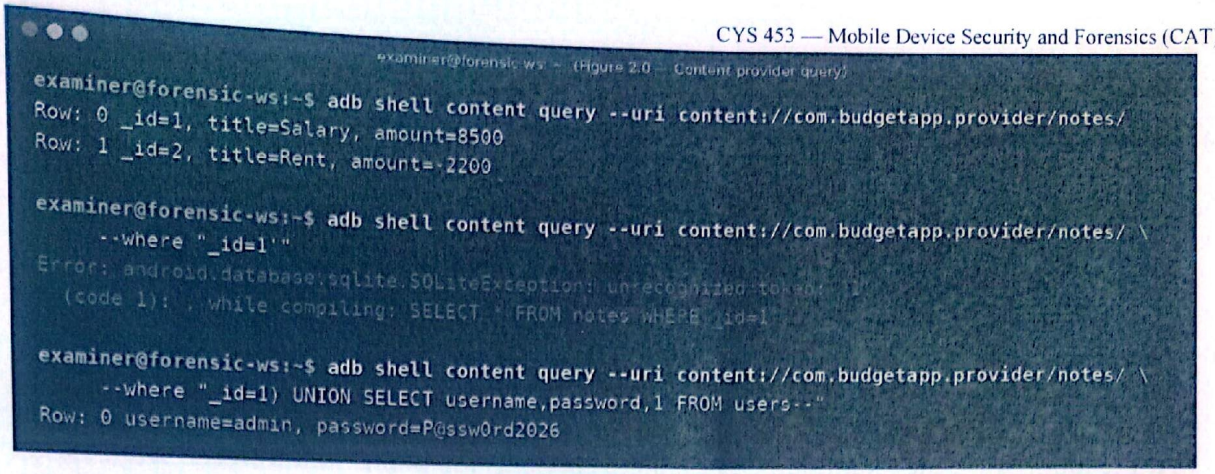


Figure 2.0

(a) Identify two (2) security weaknesses evident in Figure 2.0 and explain how each could be exploited by an attacker. **[4]**

(b) Recommend two (2) secure coding practices that would prevent this vulnerability, justifying each. **[4]**

Question 3 [7 Marks]

A junior examiner recovers a suspect's smartphone during a raid. The device is already unlocked at the time of seizure. The junior examiner argues: "Since the phone was already unlocked when we got it, there is nothing left to search for, so we do not need a warrant before examining its contents."

(a) Critically evaluate the junior examiner's statement with reference to the principles of lawful seizure and scope of authority. **[4]**

(b) State two (2) consequences that could follow from proceeding with the examination without proper legal authority. **[3]**

Question 4 [25Marks]

Read the following passage carefully, then answer the questions that follow. All answers should be supported with specific details drawn from the passage.

At 06:45, officers executing a search warrant at a residential property in Kitwe recovered a smartphone from a bedside table. The phone's screen was on, showing a lock screen with a missed-call notification from a contact saved as "Boss K". The last officer to touch the phone, Constable Mwansa, later admitted he had tapped the screen once "to see if it was

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locked", which caused the screen to time out and go dark again. He did not enter a PIN and did not restart the device.

The scene photographs, taken before any handling, show the phone's status bar displaying a Wi-Fi signal icon and a battery level of 61%. No SIM card tray was visible, as the device is fully soldered internally. The digital forensic unit was called, and the phone was placed in a Faraday bag before transport to the laboratory, still powered on.

At the laboratory, the examiner, Ms Chanda, connects the phone to a forensic workstation while it remains inside a shielded enclosure. She notes that the device has not been switched off or restarted at any point since its recovery, and that it was still displaying an ordinary lock screen — not a full-disk-encryption startup screen requesting a passcode to decrypt storage — when it was placed in the bag. Ms Chanda explains to a trainee that this single detail matters more than almost anything else in the case file, because it determines which encryption keys are still resident in the device's memory, and therefore what an examiner can realistically attempt.

Before connecting any acquisition tool, Ms Chanda first confirms that the Faraday bag's seams show no gaps, and that the phone's signal indicator — visible through a small transparent window sewn into the bag — shows no bars. Only once this is confirmed does she remove the phone from the bag, briefly, to switch on Airplane Mode herself, before returning it to a second, static-free evidence bag for the remainder of the examination.

(a) According to the passage, was the phone in a Before First Unlock (BFU) or an After First Unlock (AFU) state when it reached the laboratory? Identify two (2) specific details from the passage that support your answer. **[5]**

(b) Constable Mwansa tapped the screen once without entering a PIN, and did not restart the device. Explain, based on the passage, why this action did not change the device's encryption state, and describe one action that WOULD have changed it materially. **[4]**

(c) Identify the network-isolation method described in the passage, and explain the sequence of two checks Ms Chanda performed before removing the phone from the bag. **[4]**

(d) Explain why Ms Chanda still switches on Airplane Mode herself rather than relying on the Faraday bag alone for the rest of the examination. **[4]**

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(e) Using details from the passage, explain why the absence of a visible SIM card tray does not necessarily mean the device has no subscriber identity module to investigate. **[4]**

(f) The initial scene photographs recorded the phone's battery level as 61%. Explain why this detail is forensically relevant to how the examiner should prioritise her next steps. **[4]**

— END OF TEST —

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